
ERIK ERP – MASTER PRODUCT SPECIFICATION & TECHNICAL WHITEPAPER

(Enterprise Resource & Intelligence Kernel)

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Date: [20th October 2025]

Confidential Document – For Development, Investment, and Licensing Quotation Purposes

1. Executive Overview

ERIK ERP (Enterprise Resource & Intelligence Kernel) is a comprehensive, multi-sector enterprise resource planning system engineered to unify financial, human resource, operational, and compliance management across **private, public, and international organizations**.

The system delivers full automation of **Accounting, HR, Payroll, Procurement, Banking, Taxation, Inventory, Manufacturing, and Communication** while aligning with **Zambian statutory frameworks** (ZRA Smart Invoice, NAPSA, NHIMA, PACRA, CEEC, IFMIS) and supporting **international identity and financial compliance**.

Built as both a **national and global kernel**, ERIK ERP empowers organizations to operate seamlessly across borders, languages, and currencies while preserving full regulatory compliance and automation.

2. Vision & Purpose

ERIK ERP is designed to:

- Digitize and integrate all critical enterprise processes.
- Bridge government, business, and community ecosystems.
- Serve both **local SMEs** and **international corporations** through **SaaS, PaaS, and on-premise** models.
- Enable **smart, automated decision-making** via built-in AI.
- Support **sustainable digital transformation** in line with Zambia's Vision 2030 and Smart Zambia initiatives.

3. Core System Identity & Access Framework

3.1 Multi-Channel Registration

- **National Registration (Zambia):** NRC and Smart Zambia eID integration.
- **International Registration:** Passport, national ID, or globally verifiable digital ID (e.g., NIN, SSN, Emirates ID).
- **Corporate Registration:** TPIN, PACRA verification, and corporate email validation.
- **User IDs:** Each user receives a unique ERIK-ID (linked to NRC, Passport, or verified email).
- **Email and Mobile Verification** required for all new profiles.
- **Biometric or Face-ID login (optional)** for secured environments.

3.2 Access & Security

- Multi-factor authentication (2FA) via email/SMS.
- Role-based access control (RBAC).
- Device fingerprinting & trusted device management.
- IP and geolocation-based login control.
- Auto session timeout, logout, and audit trails.
- End-to-end encryption and full activity logs.

4. Human Resource, Payroll & Talent Ecosystem

4.1 HR Core

- Employee master database (linked to NRC/ID/passport).
- Employment contracts, grades, and departments.
- Leave, overtime, and shift management.
- Payroll engine (PAYE, NAPSA, NHIMA, WCF, gratuity, bonuses).
- Loans and salary advance management.
- Employee benefits, allowances, and deductions.
- Pension and retirement benefit tracker.
- Multi-company HR compliance (local and international).

4.2 Talent Development & Marketplace

- HR Job Advertisement Portal (public & internal).
- Integrated **CV Writer & Profile Builder** with AI suggestions.
- Recruitment pipeline and applicant tracking.
- Interview scheduling and scoring sheets.

- AI-assisted **End-of-Year Performance & Appraisal Reports**.
- AI-generated **Management Notes** for HR reports and audits.

4.3 Self-Service & Engagement

- Employee Self-Service (ESS) dashboard for leave, payslips, training, and documents.
- HR Document Generator (letters, contracts, warning notices).
- Grievance tracker, training planner, and KPI dashboard.
- Internal messaging and document upload.

5. Finance, Accounting & Compliance Engine

- Full double-entry journal engine (compact format).
- General ledger, chart of accounts, and trial balance builder.
- Multi-currency and exchange rate automation.
- Amortization, depreciation, and asset revaluation.
- Automated posting from HR, procurement, and inventory modules.
- IFRS/IAS/GAAP/IPSASB toggle for accounting standards.
- ZRA Smart Invoice and TPIN validation compliance.
- Tax pack generator for VAT, PAYE, Corporate, Withholding, Excise, Customs, and Property Tax.
- Audit-ready statements with journal drill-down.
- AI-assisted **End-of-Year Financial Reports** with **Executive Management Commentary**.
- Consolidated reports across branches, subsidiaries, and countries.

6. Banking, Payments & Financial Automation

- **Bank API Integration:** API integration with Zambian and international banks.
 - Automated debit and credit postings directly from bank feeds.
 - API connectors for major Zambian and regional banks (ABSA, Stanbic, FNB, Zanaco, Atlas Mara).
 - Real-time transaction sync.
- Automated debit and credit entry posting from live bank feeds.

Automated Posting Rules:

- Customer receipts → Credit AR
 - Supplier payments → Debit AP
 - Bank charges → Expense ledger
- Bank reconciliation engine with AI-driven auto-match.

- Mobile money API integration (MTN, Airtel, Zamtel).
 - Customer and supplier payment gateway links.
 - Petty cash, floats, and digital receipt management.
 - Teller slip and deposit register matching.
 - Secure online payment portal for clients.
 - Real-time bank dashboards and liquidity monitoring.
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7. Procurement, Inventory & Manufacturing

- Item Master with category, units, weights, cost, and stock value.
 - Supplier Management (TPIN, tax type, compliance rating).
 - Purchase Requisitions, Approvals, and Orders.
 - Goods Received Notes (GRN) and delivery matching.
 - Inventory reconciliation and valuation (FIFO, LIFO, weighted average).
 - OCR scanning for supplier invoices and purchase receipts.
 - Stock transfers (horizontal and vertical).
 - Manufacturing process flow (Raw → WIP → Finished Goods).
 - Batch costing, scrap tracking, and production rework analysis.
 - Minimum reorder level alerts and expiry notifications.
 - Barcode and QR generation for product labeling.
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8. Reporting, Analytics & Intelligence

- Real-time data visualization dashboards.
- Custom report builder with **time-grain filters** (daily, weekly, monthly, quarterly, annual, custom).
- Tax compliance reports per module.
- AI-assisted variance analysis and forecasting.
- Consolidated financial and management reports.
- Performance dashboards for executives, departments, and auditors.
- Export to PDF, Excel, CSV, or live JSON API feed.
- Scheduled email and WhatsApp report distribution.
- Department, Branch, and Consolidated Reports.
- Budget vs. Actual Reports.
- Comparative Analysis (multi-period).
- Statutory Reports: VAT, PAYE, NAPSA, NHIMA, Corporate Tax, CEEC, and CDF.
- Audit Trail and User Activity Logs.
- Interactive Dashboards with Charts & KPIs.
- Export Options: PDF, Excel, CSV, JSON, or live API feed.
- Automated Scheduled Reports (daily/weekly via email or WhatsApp).

- Management Dashboard: Revenue, Expenses, Profitability, Employee costs.
 - Compliance Report Builder: Tax health, statutory filing readiness, Smart Invoice logs.
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9. AI, Automation & Document Intelligence

- AI-driven OCR scanning for invoices, receipts, and statements.
 - Auto-entry suggestions for repetitive transactions.
 - Predictive cash flow and HR forecasting.
 - AI anomaly detection (duplicate entries, fraud).
 - Workflow automation (approvals, escalations).
 - Machine learning-driven document classification (invoice, PO, GRN).
 - Intelligent anomaly detection (duplicates, inconsistencies).
 - ERIK AI Assistant (interactive chatbot) for:
 - Accounting queries
 - Report summaries
 - Helpdesk and tutorials
 - Management commentary generation
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10. Communication & Collaboration Platform

- **Internal Chat:** Department and project-based messaging with file sharing.
 - Real-time chats between departments, projects, and teams.
 - File sharing and tagging.
 - **External Chat:** Supplier and customer communication portal.
 - Client and supplier portals with integrated messaging.
 - **Email Integration:** Automated dispatch of invoices, statements, payslips, and reports.
 - **WhatsApp Integration:** Send invoices, statements, reminders, and alerts directly through verified WhatsApp Business accounts.
 - Send invoices, statements, and reminders directly via WhatsApp Business API.
 - Real-time communication tracking in chat logs.
 - **Notifications:**
 - Push, SMS, and in-app alerts for approvals, tasks, and reminders.
 - **Announcements:** Internal newsfeed and public updates via dashboard or portal.
 - **Social Integration:** National and international ID-based login and professional profiles.
 - Company newsfeed, updates, and events.
 - Integration with national registry for verified user interactions.
 - Public feedback and compliance performance channels.
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11. Multi-Sector, Multi-Entity & Consolidation Engine

- Multi-company, multi-branch, and multi-country setup.
- Consolidated accounting, HR, and inventory across entities.
- Centralized or decentralized configuration/data control (government-compatible).
- Industry templates for:
 - Agriculture & Livestock
 - Manufacturing & Processing
 - Construction & Real Estate
 - Public Sector (IFMIS, CEEC, CDF)
 - Mining & Energy
 - Hospitality & Tourism
 - Retail & Wholesale
 - Education Institutions
 - Healthcare & Diagnostics
 - Transport, Logistics, Retail, Hospitality, Healthcare.
- Custom industry builder for emerging sectors.
- Vertical and horizontal integration across organizations, departments, and sectors.
- Consolidation of financials, HR, and operations across multiple entities.

12. Pricing & Licensing System

Tier	Target Users	Access & Features
Individual Free	Freelancers, students, individuals	Basic personal finance, reports, and learning sandbox
Individual Basic	Sole traders, consultants	Accounting + HR + invoice automation
Individual Premium	Professionals & small agencies	All individual features + Smart Invoice + AI reporting
Company Free Basic	Startups & NGOs (≤3 users)	Basic accounting, payroll, and procurement
Company Premium	SMEs (≤10 users)	Full HR, accounting, and inventory modules
Enterprise	Mid to large firms	All modules + bank APIs, OCR, Smart Invoice
Agent	Corporates, government, and NGOs	Multi-branch + consolidation + integrations
	Distributors, resellers, consultants	Multi-client access, commissions, and analytics dashboard

Licensing Features

- Role-based license validation.
- Auto-renewal and upgrade options.
- Watermark control (disabled for Premium & Enterprise).
- Offline validation for on-premise installations.

- AI-monitored license analytics for usage tracking.

12.2 Licensing Controls

- Auto-license activation via online validation.
 - Role-based feature unlock by tier.
 - Subscription and lifetime options.
 - Watermark control (disabled in Premium & Enterprise).
 - Auto-scaling for cloud-hosted tenants.
 - Free-to-paid upgrade path with migration tool.
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13. Deployment & Infrastructure Models

13.1 SaaS (Software as a Service)

- Cloud-hosted on secure regional data centers.
- Automatic updates, patching, and scaling.
- Subscription-based pricing.
- Ideal for SMEs, startups, and distributed organizations.

13.2 PaaS (Platform as a Service)

- Hosted platform with client-managed applications.
- White-labeled for government or large enterprise resellers.
- Full API access for integration with custom systems.

13.3 On-Premise / Hybrid Model

- Installed on client-managed servers.
 - Recommended for banks, government, and sensitive institutions.
 - Periodic synchronization with ERIK Cloud for analytics and AI updates.
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14. Marketplace & Ecosystem

- **ERP Marketplace Portal** where users can:
 - Purchase add-ons and modules.
 - Subscribe to HR services (job ads, CV writing, outsourcing).
 - Hire accountants, consultants, and system integrators.
- **HR Marketplace:**

- Job advertisement posting.
 - Verified recruiter and agency profiles.
 - CV & cover letter writing powered by AI.
 - **Report Marketplace:**
 - AI-assisted financial, HR, and management report templates.
 - End-of-Year report writers with integrated analytics and AI commentary.
 - **Agent Portal:**
 - Revenue tracking and commission dashboards for partners.
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15. Compliance, Legal & Governance

- Data protection aligned with Zambian ICT and privacy laws.
 - GDPR-style compliance for international clients.
 - Automated data retention and archival.
 - Legal disclaimer and system usage policy templates.
 - Role-based data access restriction.
 - Audit log export for regulators and auditors.
 - Integration with IFMIS, PACRA, NHIMA, NAPSA, and CEEC APIs.
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16. Sustainability & Green Initiative

- “Green Check” compliance dashboard tracking environmental responsibility.
 - Resource usage and energy tracking reports.
 - Eco suggestion system with employee engagement.
 - Optional public certification badge for sustainable companies.
 - Eco-awareness and sustainability tracking.
 - Employee and company green challenges.
 - Carbon footprint estimates.
 - Green Check Accreditation score for compliance and CSR reporting.
 - Optional public listing for compliant organizations.
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17. Administration & Control Features

- Centralized admin dashboard for user, license, and access management.
- Workflow builder for approvals and escalations.
- Global configuration for taxes, currencies, and permissions.
- Role-based user interface personalization.
- Backup, restore, and version control modules.
- Licensing and Subscription Management.

- Audit Trail Viewer.
 - Workflow Builder (custom approval chains).
 - Compliance Policy Editor.
 - IP & Trademark Protection Log.
 - Legal Document Generator (NDAs, SLAs, Contracts).
 - Data Retention & Privacy Configuration (GDPR-style).
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18. Developer, API & Integration Toolkit

- RESTful & GraphQL APIs for third-party systems.
 - Webhooks for event-based triggers (real-time updates).
 - Partner SDK for third-party developers, government and educational institutions.
 - Import/export tools for CSV, Excel, XML, JSON.
 - Plugin builder and integration testing environment. Plugin Store for modular extensions.
 - Sandbox for development partners.
 - Bulk import/export tools (CSV, Excel, XML, JSON).
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19. Mobile & Multi-Device Access

- Native mobile apps for Android & iOS.
 - ESS and manager access from mobile dashboards.
 - Push notifications for alerts and approvals.
 - QR login for remote workers.
 - Geo-tagging for attendance and deliveries.
 - Offline mode with background synchronization.
 - Biometric unlock for mobile users.
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20. Hosting, Infrastructure & Integration

- Cloud-first with hybrid (on-prem + cloud) support.
- Daily encrypted backups and redundancy.
- Integration with local data centers (e.g., Paratus, Liquid, MTN).
- API Gateway for external apps and public institutions.
- Compatibility with IFMIS, ZRA Smart Invoice, NHIMA, NAPSA, CEEC, and PACRA APIs.
- Continuous deployment (CI/CD) pipeline.
- AI monitoring for uptime, load, and threat detection.

21. Future Expansion Roadmap

- E-commerce & POS integration (ERP-Commerce).
- Blockchain-based audit trails.
- AI-driven chatbot for clients and support teams. Interface for accounting & HR tasks.
- Predictive procurement and supply chain optimization.
- Integration with regional central banks and trade platforms.
- Multi-language support (English, French, Swahili, Arabic).
- Smart Contract Management for supplier automation.
- Global currency and tax extension for multi-nation operations.

22. Intelligence & Forecasting

- Predictive analytics for cash flow, HR costs, and sales trends.
- Performance dashboards per module.
- Anomaly and fraud detection alerts.
- Budget recommendation engine.
- Automated reminders and compliance alerts.

23. Conclusion

ERIK ERP stands as a next-generation platform designed for national scalability and global interoperability.

It unites **finance, HR, operations, and intelligence** into a single, secure, and smart system capable of driving digital transformation across all sectors.

Through its AI-driven architecture, industry templates, flexible deployment models, and integrated marketplace, ERIK ERP provides the foundation for sustainable growth, accountability, and automation across Africa and beyond.

📄 ERP Product Specification Document

For Request for Quotation (RFQ) – Development, Training, Licensing, Hosting, Support, Bug Fixes, Consultations & Reviews

Client: Greenomic Enterprise Limited (Rose Green Lines Farms Limited subsidiary)

ERP Name: *ERIK ERP (ERIK – Enterprise Resource & Intelligence Kernel)*

Date: [20th October 2025]

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1. 🚀 Introduction & Purpose

Greenomic Enterprise seeks quotations from qualified Zambian software development companies to **develop, deploy, and support** a modular ERP system that covers **Accounting, HR & Payroll, Inventory & Procurement, Compliance, Banking, Manufacturing, Public Sector, Agriculture, and Multi-Sector Integrations**.

The ERP must:

- Meet **Zambian statutory requirements** (ZRA Smart Invoice, TPIN, NAPSA, NHIMA, CEEC, PACRA, IFMIS).
 - Be **modular, scalable, and multi-tenant** (SMEs, corporates, public institutions).
 - Support **offline/online sync**.
 - Provide **multi-currency & exchange rate handling**.
 - Include **training, licensing, hosting, bug-fix support, and consultations**.
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2. 🚀 Scope of Work

2.1 Core ERP Modules (Phase 1–40+ Roadmap)

1. Authentication & Security

- 2FA, Role-based Access, Audit Logs, Device Fingerprinting, Whitelisting.

2. Finance & Accounting

- General Ledger, Trial Balance, Chart of Accounts, Journal Engine (dual-entry compact format), Tax Pack Builder, Petty Cash, Bank Reconciliation, Revenue Collection, ZRA Smart Invoice Connector.

3. HR & Payroll

- Employee Master Data, Leave, Payroll, Payslips, Loans & Advances, Overtime, Grievances, Employee Separation, KPI & SLA Monitoring, Pension & Retirement, Allowances, Welfare, Casual Workers.
4. **Inventory & Procurement**
 - Item Master, Supplier Management, Purchase Orders, GRN & Delivery Matching, Stock Reconciliation, Inventory Valuation, Purchase Returns, OCR-enabled Receipt & Invoice Scanning.
 5. **Operations & Production**
 - Manufacturing inputs/outputs, transfer pricing, cost centers, profit centers, batch processing.
 6. **Compliance & Taxation**
 - VAT, PAYE, Corporate Tax, Customs, Excise, Property Tax, Withholding Tax, ZRA Smart Invoice mapping, JSON exports, CDF/CEEC reporting.
 7. **Public Sector & Government Add-ons**
 - IFMIS integration, CDF tracking, audit-ready reports, centralized & decentralized operations.
 8. **Other Modules**
 - Employee Self-Service, Job Requisitions, Document Generation, Performance Reviews, Training & Development, Green Initiative Module.
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2.2 Technical Requirements

- **Backend:** Python (FastAPI), PostgreSQL with SQLAlchemy + Alembic migrations.
- **Frontend:** Dual approach:
 - **Web stack:** HTML, CSS, JS.
 - **Python-based:** Streamlit / FastAPI templating for Replit, Heroku, Vercel, Render, Railway.
- **Hosting:** Cloud (AWS, Azure, GCP, or Zambian hosting provider) + hybrid options.
- **Security:**
 - SSL/TLS encryption.
 - Role-based access.
 - Audit trail for every transaction.
 - Anti-virus & cyber-attack monitoring.

- **Compliance:** ZRA Smart Invoice integration, TPIN validation, NHIMA/NAPSA deductions, IFRS/IAS/GAAP toggle.
 - **Scalability:** Must handle 1–50,000+ users with high availability (HA) and load balancing.
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2.3 Deliverables

1. **ERP System** – fully developed, tested, and deployed.
 2. **Training Program** – Admin, Developer, Client, HR, Government, and Technical manuals.
 3. **Licensing System** – Tiered plans (Free, Premium, 10-year license, Lifetime license).
 4. **Hosting & Infrastructure** – Deployment setup, backup policies, disaster recovery.
 5. **Support Package** – Bug fixes, consultations, periodic reviews.
 6. **Custom Reporting Suite** – Daily, weekly, monthly, annual reports (customizable).
 7. **User Guides & Documentation** – PDFs, Word docs, video simulations.
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3. ★ Training Requirements

Vendors must provide:

- **Onsite & Virtual Training** – for staff, administrators, and clients.
 - **Training Simulations** – ERP sandbox environment (with HR, Payroll, Finance, Inventory).
 - **Learning Integration (GCLIP)** – Student/educator onboarding with certification.
 - **Manuals Required:**
 - Developer manual
 - Client manual
 - HR manual
 - Admin manual
 - Technical manual
 - Creator's manual
 - Integrity Oath (for developers, admins, government users)
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4. ★ Licensing & Monetization

Vendors must include:

- **License Management System** – Free, Premium, Lifetime (10 years).
 - **Watermarking** – Default in documents, optional removal for premium users.
 - **Tiered Pricing** – Enterprise, SME, Government.
 - **Royalty model** – Configurable.
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5. 🚀 Hosting & Infrastructure

- **Primary Option:** Cloud hosting in Zambia or global (AWS, Azure, GCP).
 - **Secondary Option:** Hybrid (cloud + local physical server).
 - **Backup & Recovery:** Daily backup, restore tests, legal compliance.
 - **Data Storage Policy:** Cannot fall below legal minimums; client must maintain local backup.
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6. 🚀 Support & Maintenance

- **Bug Fixes:** Must be addressed within SLA timelines.
 - **Consultations:** Scheduled quarterly & ad-hoc.
 - **System Reviews:** Bi-annual with audit logs.
 - **Upgrades:** Security patches, new modules, Smart Invoice compliance updates.
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7. 🚀 Roles & Permissions

- **Admins** – Full access.
 - **Managers** – Department-specific access.
 - **Employees** – ESS portal access.
 - **Auditors/Government** – Read-only, drill-down views.
 - **Developers** – With NDA & IP restrictions.
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8. 🚀 Vendor Response Requirements

Vendors must include in quotation:

1. **Company Profile** (experience in ERP, local government integrations).
2. **Proposed Architecture** (deployment diagram, tools, hosting).

3. **Cost Breakdown:**

- Development & customizations.
- Training.
- Licensing setup.
- Hosting.
- Support & bug fixes.
- Consultations & reviews.

4. **Timeline** (phase-based implementation).

5. **Post-Go-Live Support Model.**

6. **IP/Confidentiality Agreement** (Greenomic holds full IP rights).

9. **Evaluation Criteria**

Quotations will be evaluated based on:

- Compliance with technical specs.
 - Local Zambian ERP/government integration experience.
 - Total cost of ownership (development + support).
 - Training & documentation quality.
 - SLA & bug-fix guarantees.
 - Security & compliance readiness.
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